

Junmo Cho

Ph.D. Student, Korea Advanced Institute of Science and Technology (KAIST)

Github: , Twitter: , LinkedIn: , Google Scholar: 

E-mail: junmokane@kaist.ac.kr

Summary

I am a Ph.D. researcher at KAIST working on embodied agents, spatially aware intelligence, and robotics.

My earlier research focused on building embodied agents with long-term spatial memory that can remember where they have been and what events occurred in open-ended environments such as Minecraft, as well as on incorporating spatial structure directly into transformer architectures, and showed its spatial reasoning abilities (Spatially-Aware Transformer).

Recently, my research has shifted toward robotics, with a particular interest in bridging real-world environments and physics-based simulators. I am exploring real2sim2real pipelines that reconstruct real environments in simulation and learn task policies using video-based world models and vision-language-action(VLA) models, with the goal of robust deployment in real-world robotic systems.

Education

KAIST

Ph.D. in Graduate School of AI

Advisor: Prof. Sungju Hwang

GPA: 4.11 / 4.3

Seoul, Korea

Aug. 2022 – Present

KAIST

M.S. in Electrical Engineering

Advisor: Prof. Young-Gyu Yoon

GPA: 4.11 / 4.3

Daejeon, Korea

Feb. 2022

KAIST

B.S. in Electrical Engineering and Computer Science (Double Major)

Minor: Mathematical Science

GPA: 3.99 / 4.3

Daejeon, Korea

Feb. 2020

Publication

[C6] T. Nam, **J. Cho**, Y. Jang, S. Hwang. SpeedAug: Policy Acceleration via Tempo-Enriched Policy and RL Fine-Tuning, *Under Submission*, 2026.

[C5] **J. Cho***, S. Kim*, S. An*, M. Kim, D. Lee, H. Lee, S. Hwang, H. Lee. GFlowPO: Generative Flow Network as a Language Model Prompt Optimizer, *Under Submission*, 2026.

[C4] J. Park*, **J. Cho***, S. Ahn. Mr. Steve: Instruction-Following Agents in Minecraft with What-Where-When Memory, *International Conference on Learning Representations (ICLR)*, 2025.

[C3] **J. Cho***, J. Yoon*, S. Ahn. Spatially-Aware Transformer for Embodied Agents, *International Conference on Learning Representations (ICLR)*, 2024, **Spotlight**.

[C2] **J. Cho***, S. Han*, K. Shin, Y.-G. Yoon. Robust and Efficient Image Alignment of Biomedical Images through Simultaneous Low Rank and Sparse Decomposition,

Winter Conference on Applications of Computer Vision (**WACV**), 2023.

[C1] **J. Cho**, D.-H. Lee, Y.-G. Yoon. Inducing Functions through Reinforcement Learning without Task Specification, *NeurIPS Workshop Deep Reinforcement Learning (NeurIPS Workshop Deep RL)*, 2022.

[J1] W. La*, S. Bae*, J. Seo*, H. Yu*, **J. Cho**, H. Kim, H. Nam, S. Han, E. Yi, E. Kim, C. Kang, H. Shin, C. W. Song, Y.-G. Yoon, J.-B. Chang. Abunmix enables the simple and robust multiplexed immunofluorescence imaging, **VIEW**, 2026.

Experience

MLAI, KAIST

Seoul, Korea

Ph.D. Student

Sep. 2025 -

- Conducted research on GFlowNet-based Prompt Optimization method. Proposed GFlowPO which finetunes LLM with GFlowNet and training-free Dynamic Memory Update, outperforms previous prompt optimization baselines on reasoning benchmarks.
- Conducted research on accelerating diffusion-based policies using learning-based methods. Demonstrated that augmenting demonstration data with randomized temporal downsampling and fine-tuning with reinforcement learning achieves faster execution and higher success rates than heuristic acceleration methods.

MLMLab, KAIST

Daejeon, Korea

Ph.D. Student (Prev.)

Mar. 2022 - Aug. 2025

- Conducted research on spatial memory of transformer. Showed spatially-aware transformer can spatially reason and memorize outperforming transformer on spatial tasks.
- Conducted research on memory architecture for open-world embodied agent in Minecraft. Proposed Place-Event memory for Minecraft agent outperforming on instruction following sequential tasks.

NICA Lab, KAIST

Daejeon, Korea

M.S. Student

Mar. 2020 - Feb. 2022

- Conducted research on memory component in RL setting. Implemented baselines (recurrent dqn, R2D2) for comparison. Also modeled specific POMDP and showed our algorithm performed best.
- Conducted research on biomedical image alignment (Zebrafish brain, mouse brain, x-ray images). Proposed fast and scalable algorithm REALS.
- Conducted research on blind source unmixing on fluorescence microscopy images.

NICA Lab, KAIST

Daejeon, Korea

Research Assistant

Aug. 2019 - Dec. 2019

- Implemented RNN based neural tracing on Brainbow image with ground truth neural trace available.
- Conducted research on defending adversarial attack using GAN and genetic algorithm.

KAIST Institute (KI)

Daejeon, Korea

Research Assistant

Feb. 2019 - Dec. 2019

- Conducted research on noisy sound localization, and participated in 2019 AI Grand Challenge sound recognition track.
- Participated in AutoDL challenges (Auto Speech) and designed general audio classification (Speaker, Emotion, Accent, Language, Music genre) model.

Honors and Awards	Un Chong-Kwan Scholarship Award, School of Electrical Engineering, KAIST	2020
	· Achievement of excellence in 2020 entrance examination	
	EE Fellowship, School of Electical Engineering, KAIST	2020
	· Honor for outstanding students among freshmen of Master's course	
	Yoon & Kim Research Scholarship, KAIST Brain Science Research Center	2019
	· Selected for research scholarship project for undergraduate students	
Teaching Experience	The 8th Big Data Analysis Competition (UNIST) Best Award	2019
	· Awarded for proposing business model which predicts liquid seaborne	
	National Excellence Scholarship, Korea Student Aid Foundation	2015-2018
	· Honor for outstanding students among freshmen of undergraduate course	
	Dean's list (2015 Spring, 2018 Fall)	
	· Academic award for top 3 percent students	
Technical Skills	Teaching assistant for CS672 in KAIST	Fall 2024
	· Managed the course 'Reinforcement Learning'.	
	Teaching assistant for CS492 in KAIST	Spring 2024
	· Managed the course 'Special Topics in Computer Science<Deep Reinforcement Learning and Game AI>'.	
	· Won Best TA Award.	
	Teaching assistant for CS672 in KAIST	Fall 2023
	· Managed the course 'Reinforcement Learning'.	
	Teaching assistant for CS492 in KAIST	Spring 2023
Activities (Extracurricula)	· Managed the course 'Special Topics in Computer Science<Deep Reinforcement Learning>'.	
	Teaching assistant for CoE202 in KAIST	Fall 2020
	· Managed the course 'Basics of Artificial Intelligence <Big data analysis and machine learning>'.	
	Freshmen Tutoring Program in KAIST	Spring 2019
	· Helped freshmen student(tutee) to study 'Calculus I'(MAS101) in KAIST.	
	Mathematical skills Real Analysis, Numerical Analysis, Linear Algebra, Convex Optimization, Probability Theory, Stochastic Process, Bayesian Statistics, Lebesgue Integral Theory	
	Programming languages Python, MATLAB, C/C++	
	Frameworks Pytorch, IsaacSim, Mujoco, Tensorflow	
	Languages Korean(Native), English(Fluent)	
Activities (Extracurricula)	KAIST Educational Outreach Club SEED	Daejeon, Korea
	President (Founder) & Member	Dec. 2016 - Aug. 2018

- Conducted and participated in educational outreach program for multi-cultural students in Daejeon [\[Web\]](#).

KAIST Formula Club **ZILZU**

Daejeon, Korea

Member

Mar. 2016 - Aug. 2016

- Learned grinding, wiring and participated in building off-road type car.
- Participated in 2016 KSAE Baja/Formula/EV in Baja part(Off-Road Racing).

Address

108, Taebong-ro, Seocho-gu, Seoul, Republic of Korea